

Trigonometric ratios



1

a $\frac{5}{13}$

b $\frac{15}{17}$

2

a $\frac{4}{5}$

b $\frac{24}{25}$

3

a $\angle BCA$

b $\frac{3}{5}$

c $\frac{4}{3}$

4

a $\angle BCA$

b $\frac{4}{5}$

c $\frac{4}{3}$

5

a $\angle BCA$

b $\frac{8}{17}$

c $\frac{15}{17}$

6

a $x = 13$

b $\frac{12}{13}$

c $\frac{5}{13}$

7

a $x = 10$

b $\frac{4}{5}$

c $\frac{3}{5}$

8

a $x = 4$

b $\frac{4}{3}$

9

a $AC = 12$ units

b $\frac{12}{5}$

10

a $\frac{24}{25}$

b $\frac{24}{25}$

c $\cos(90^\circ - \theta) = \sin \theta$

11

a $\frac{3}{5}$

c $\sin(90^\circ - \theta) = \cos \theta$



b $\frac{3}{5}$

12

a $\frac{12}{5}$

b $\frac{12}{5}$

c $\tan \theta = \frac{\sin \theta}{\cos \theta}$

Reciprocal trigonometric ratios

13 $\frac{5}{4}$

14 $\frac{5}{4}$

15 $\frac{7}{24}$

16

a $\angle BCA$

b $x = 3$

c $\frac{5}{3}$

d $\frac{3}{4}$

17

a $\angle BCA$

b $x = 17$

c $\frac{17}{8}$

d $\frac{17}{15}$

18

a 17 units

b

i $\frac{8}{17}$	ii $\frac{15}{17}$	iii $\frac{15}{8}$	iv $\frac{17}{8}$
v $\frac{17}{15}$	vi $\frac{8}{15}$		

19

a

i $\frac{AC}{BC}$

ii $\frac{BC}{AB}$

b

i $\frac{AC}{AB}$

ii $\tan \alpha = \frac{BC}{AC}$

c

i $\frac{BC}{AB}$

ii $\frac{AC}{AB}$

d

i $\frac{8}{15}$

ii $\frac{15}{17}$

e

i $\frac{15}{17}$

ii $\frac{15}{8}$

f

i $\frac{24}{25}$

ii $\frac{24}{25}$

20

a 9 units

b

i $\frac{4}{5}$

ii $\frac{3}{5}$

iii $\frac{4}{3}$

iv $\frac{5}{3}$

v $\frac{5}{4}$

vi $\frac{3}{4}$

21

a $\frac{4}{5}$

b $\frac{15}{8}$

c $\frac{5}{13}$

d $\frac{5}{4}$

e $\frac{4}{7}$

f $\frac{13}{5}$